

Pretalk remark: E_k -monoidal n -category $\rightsquigarrow$ $(n+k)$ -fold decorated n -manifolds
 "string calculus"

Functoriality of Quantum Dimension (η -trace and dimension) aimed towards finding an invariant that can distinguish S and T .

$S := 2\text{Vec}^c[\mathcal{G}]$ $T := 2\text{Vec}^r[\mathcal{G}]$
 of the 1-groupoid $\mathcal{G} = K(\mathbb{Z}_2, 1) \times K(\mathbb{Z}_2, 0)$ with braided monoidal structure determined by the delooping $B^2\mathcal{G} = K(\mathbb{Z}_2, 3) \times K(\mathbb{Z}_2, 2)$ and the following twists in $H^3(B^2\mathcal{G}; \mathbb{k}^\times)$:

Here t_n is the generator (over the Steenrod algebra) of $H^*(K(\mathbb{Z}_2, n); \mathbb{Z}_2)$, and we have used the Künneth formula and the map $t \mapsto (-1)^t$ on coefficients to map

$$H^3(K(\mathbb{Z}_2, 3); \mathbb{Z}_2) \oplus (H^3(K(\mathbb{Z}_2, 3); \mathbb{Z}_2) \otimes H^2(K(\mathbb{Z}_2, 2); \mathbb{Z}_2)) \oplus H^3(K(\mathbb{Z}_2, 2); \mathbb{Z}_2) \xrightarrow{\text{Künneth}} H^3(B^2\mathcal{G}; \mathbb{Z}_2) \xrightarrow{(-1)^t} H^3(B^2\mathcal{G}; \mathbb{k}^\times)$$

§1. 1-categorical

Let $\mathcal{E}$ be a braided monoidal 1-category, for each dualizable object $x \in \mathcal{E}$,

def. η -dimension $:= \text{ev}_x \cdot \beta_{x^*, x} \cdot \text{coev}_x = \bigcirc_x \in \Omega \mathcal{E}$

$$\text{ev}_x: x \otimes x^* \rightarrow I$$

$$\text{coev}_x: I \rightarrow x^* \otimes x$$

Claim (i) $\eta(x)$ is independent of the choice of right duality data $x^*, \text{ev}_x, \text{coev}_x$.

Proof: use naturality of β in the first component w.r.t. the unique isomorphism between choices of duality data.

(ii) if $x \cong y$ then $\eta(x) \cong \eta(y)$ [depends only on isomorphism class]

Therefore, $\eta: \pi_0 \mathcal{E}^{\text{f.d.}} \rightarrow \Omega \mathcal{E}$

Remarks (i) $\eta(x) = \eta(x^*)$

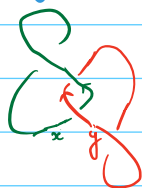
(ii) η is not monoidal in general.

for invertible x and y ,

$$\eta(x \otimes y) = \langle \beta_{y, x} \beta_{x, y} \rangle \eta(x) \eta(y)$$



pull them apart



$\in \Omega \mathcal{E}$ defined as

$$\bigcirc_x = \langle \beta_{y, x} \beta_{x, y} \rangle \uparrow \uparrow$$

def. η -trace: for a 1-morphism $\phi: x \rightarrow x$

$$\text{tr}(\phi) := \text{ev}_x \cdot (\phi \otimes \text{id}_x) \cdot \beta_{x^*, x} \cdot \text{coev}_x = \bigcirc_x^\phi \in \Omega \mathcal{E}$$

Comments: η comes from Hopf map
 for sym $\otimes$ 1-cat $\rightsquigarrow$ TFT^{fr} where $Z(*) = \mathcal{E}$, $\eta(x) = Z(S^1)$
 and remark (ii) captures orientation reversing diffeomorphism of the circle.

§2. Review of duality data for 2-categories

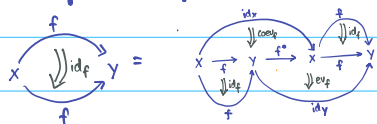
def. A right adjoint of a 1-morphism $f: X \rightarrow Y$ in a 2-category is

- a 1-morphism $f^*: Y \rightarrow X$, along with
- 2-morphisms $ev_f: f \circ f^* \rightarrow id_Y$
 $coev_f: id_X \rightarrow f^* \circ f$

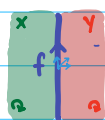
satisfying

$$id_f = (ev_f \circ id_f) \cdot (id_f \circ coev_f)$$

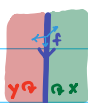
$$id_{f^*} = (id_{f^*} \circ ev_f) \cdot (coev_f \circ id_{f^*})$$



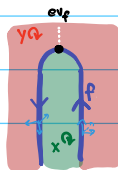
diagrammatically



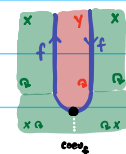
duality data:



and



,



satisfying



=



and



=

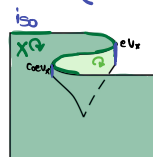


If it exists, right adjoint is unique upto unique isomorphism. (contractible choice)

defn.

A right dual of an object X in a monoidal 2-category is

- an object X^v
- 1-morphisms $ev_X: X \boxtimes X^v \rightarrow I$ and $coev_X: I \rightarrow X^v \boxtimes X$
- 2-morphism $cusp_X: id_X \xrightarrow{\cong} (ev_X \boxtimes id_X) \circ (id_X \boxtimes coev_X)$



such that $(id_{X^v} \boxtimes ev_X) \circ (coev_X \boxtimes id_X) \cong id_{X^v}$



equivalently you could ask for $cusp_X$, $cusp^{X^*}$ and compatibility

defn : An object is fully (2-) dualizable when it admits both $X^\vee$ and ${}^\vee X$ and all four ev and coev 1-morphisms admit adjoints

defn

Mates

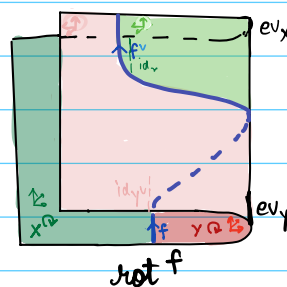
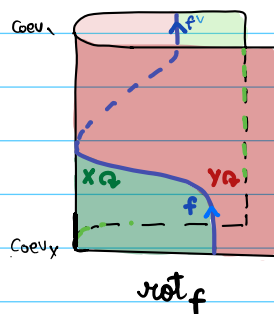
A right mate of a 1-morphism $f: X \rightarrow Y$ equipped with right duality data $(X^\vee, \text{ev}_X, \text{coev}_X, \text{cusp}_X)$ and $(Y^\vee, \text{ev}_Y, \text{coev}_Y, \text{cusp}_Y)$ is

- a 1-morphism $f^\vee: Y^\vee \rightarrow X^\vee$
- 2-morphisms

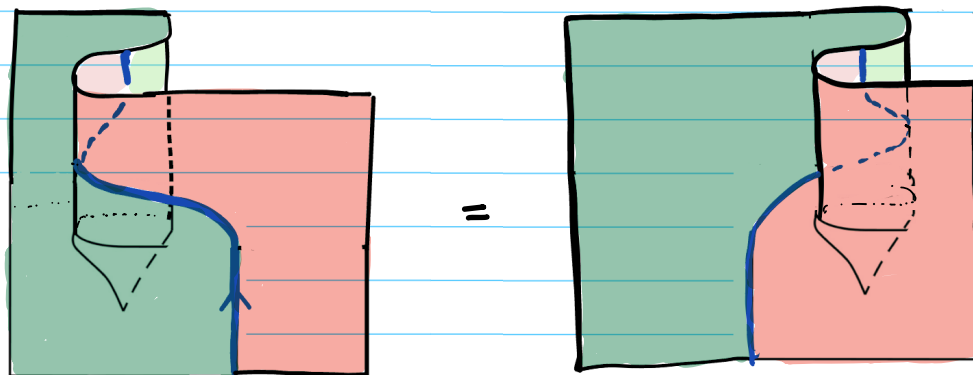
$$\text{rot}_f: (\text{id}_{X^\vee} \otimes f) \circ \text{coev}_X \xrightarrow{\sim} (f^\vee \otimes \text{id}_Y) \circ \text{coev}_Y$$

$$\text{rot}^f: \text{ev}_Y \circ (f \otimes \text{id}_{Y^\vee}) \xrightarrow{\sim} \text{ev}_X \circ (\text{id}_X \otimes f^\vee)$$

which are compatible with cusp_X and cusp_Y .



such that

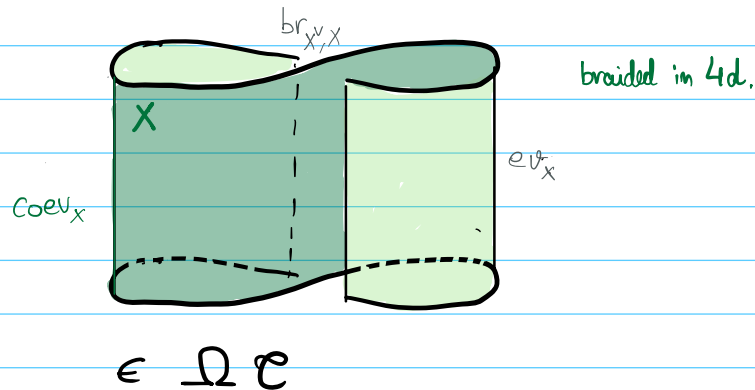


For $f: X \rightarrow Y$ and for any choice of duality data, the mate of f is unique up to unique isomorphism.

defn 'categorified' η -dimension

let $x \in \mathcal{C}$ and choose right duality data $(x^v, ev_x, coev_x, cusp_x)$.

$$\eta(x, x^v, ev_x, coev_x, cusp_x) := ev_x \circ br_{x^v, x} \circ coev_x$$



Note: isomorphism class of $\eta(x)$ only depends on equivalence class of x .

Next, we will promote η to a functor

defn 2-category of coherent dual-pairs in $\mathcal{C}$

obj: $x \in \mathcal{C}$ with choice of $(x^v, ev_x, coev_x, cusp_x)$

1-mor: $(x, x^v, \dots) \rightarrow (y, y^v, \dots)$

is $\neq$ 1-mor $f: x \rightarrow y$

* choice of right adjoint $(f^*, ev_f, coev_f)$

* choice of mates (f^v, rot_f, rot^f)

$(f^{*v}, rot_{f^*}, rot^{f^*})$

2-mor: $(f, f^v, f^*, \dots) \Rightarrow (g, g^v, g^*, \dots)$

is $\neq$ 2-mor $\phi: f \Rightarrow g$

This canonically selects 2-morphisms

$$\left. \begin{aligned} \phi^* &: f^* \Rightarrow g^* \\ \phi^v &: f^v \Rightarrow g^v \\ \phi^{*v} &: g^{*v} \Rightarrow f^{*v} \end{aligned} \right\}$$

* If all objects and 1-morphisms have duals and adjoints,

$(\mathcal{C}^{\text{coh}} \rightarrow \mathcal{C})$ is an equivalence,
 $(x, x^\vee, \dots) \mapsto x$

(fib of this map is contractible)
 (duality data)

def. The homotopy 1-category $h_1 \mathcal{C}$ of a (braided monoidal) 2-category $\mathcal{C}$ is the (braided monoidal) 1-category
 objects: objects of $\mathcal{C}$
 mor: isomorphism classes of morphisms of $\mathcal{C}$

Therefore,
 $h_1 \mathcal{C}^{\text{coh}}$

obj: $X \in \mathcal{C}$ with choice of duality data $(X^\vee, \text{ev}_X, \text{coev}_X, \text{cusp}_X)$
 1-mor: isomorphism class $[f]$ of a 1-morphism $f: X \rightarrow Y$ in $\mathcal{C}$
 (mates, adjoints uniquely determined)

$h_1 \mathcal{C}^{\text{coh}} \rightarrow h_1 \mathcal{C}$

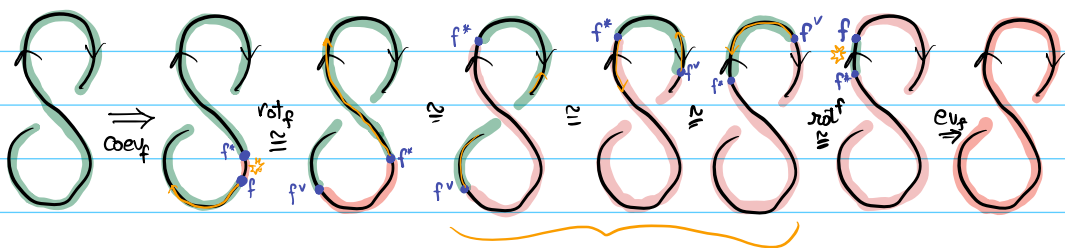
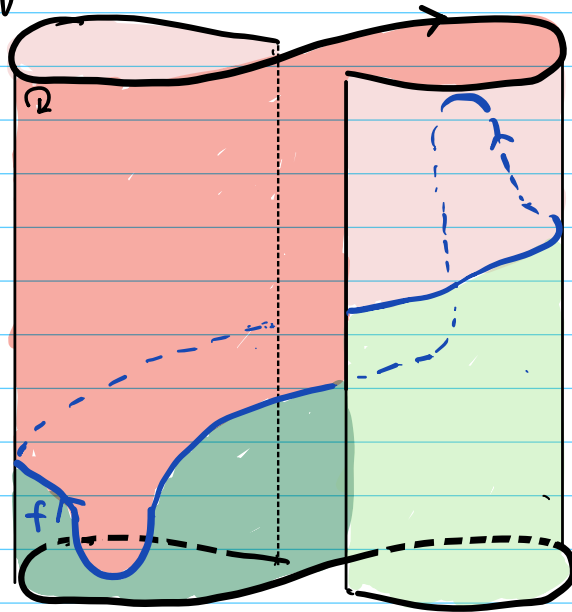
choose right duality data $\forall$ objs.

Lemma: η extends to functor $h_1 \mathcal{C}^{\text{coh}} \rightarrow \Omega \mathcal{C}$,
hence, after choosing duality data for every object,
 $\eta: h_1 \mathcal{C} \rightarrow \Omega \mathcal{C}$

Pf. on objects : $ev_X \circ br_{X^V, X} \circ coev_X$

for 1-morphism $f: X \rightarrow Y$ be a representative of
 $[f]: (x, x^V, \dots) \rightarrow (y, y^V, \dots)$ in $h_1 \mathcal{C}^{\text{coh}}$.
choose $(f^*, ev_f, coev_f)$ and (f^V, rot_f, rot_f^V)

define



Check: $\eta(f)$ independent of choice of adjoint and mate
and η assemble into a functor.

* given a 1-morphism $x: I \rightarrow I$ (choosing trivial duality data $I^V = I, \dots$)
 $\eta(x)$ agrees with $\eta(x)$ computed for $\Omega \mathcal{C}$.

* use f^* and $(f^*)^\vee$
(note $(f^*)^\vee = \Delta^*(f^\vee)$)

we can define $\tilde{\beta}_{a,b} : \text{br}_{A',B'} \circ (a \boxtimes b) \Rightarrow (b \boxtimes a) \circ \text{br}_{A,B}$

Example: let $\alpha: I \rightarrow I$ such that α, γ are both
 $\text{id}_\gamma: \gamma \rightarrow \gamma$ invertible


 $= \langle br_{idy,x} \ br_{z,idy} \rangle$


$$\eta(\underbrace{x}_{\text{green}}) = \eta(x)$$

Sketch: Consider a 2-category $\text{Coh } \mathcal{C}^{\text{Coh}}$

obj: $X + \text{left/right duals}$

1-mon: $f: X \rightarrow Y$, left mono, left adjoint
right mono, right adjoint

2-mon: $\varphi: f \Rightarrow g$

$$\begin{array}{ccccc}
 h_1(\text{coh } \mathcal{C}^{\text{coh}}) & \longrightarrow & h_1 \mathcal{C}^{\text{coh}} & (Y, Y^v, \dots) & ([g], \dots) \\
 \downarrow & & \downarrow & \downarrow & \downarrow \\
 h_1 \mathcal{C}^{\text{coh}} & \longrightarrow & h_1 \mathcal{C} & Y^v & [g^{*v}] \\
 (X, X^v, \dots) & \longmapsto & X & & \\
 ([f], \dots) & \longmapsto & [f] & &
 \end{array}$$

$$\begin{array}{ccc} (X, X^v, {}^vX, \dots) & \xrightarrow{\quad} & (X, X^v, \dots) \\ [f] & \xrightarrow{\quad} & [f] \end{array}$$

$$h_1(\text{coh } \mathcal{C}^{\text{coh}}) \xrightarrow[\textcircled{2}]{\textcircled{1}} h_1(\mathcal{C}^{\text{coh}}) \xrightarrow{\eta} \Omega \mathcal{C}$$

$$\begin{array}{ccc} (X, X^v, {}^vX, \dots) & \xrightarrow{\quad} & ({}^vX, X, \dots) \\ [f] & \xrightarrow{\quad} & [*^v f] \end{array}$$

$$\eta \circ \textcircled{2} \Rightarrow \eta \circ \textcircled{1}$$

def fl_X is the following isomorphism.

$$\eta({}^vX)$$

21 trivially self-mate as endomorphism of unit object

$$(\eta({}^vX))^v = (ev_X \circ br_{X, {}^vX} \circ coev_X)^v$$

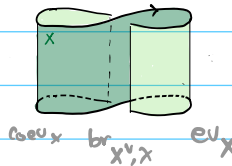
21 $()^v$ is contravariant

$$(coev_X)^v \circ (br_{X, {}^vX})^v \circ (ev_X)^v$$

21 observations

$$ev_X \circ br_{X, {}^vX} \circ coev_X$$

$$\eta''(X)$$



+ naturality

for $f: X \rightarrow Y$

$$\text{we find } \eta(f) \cdot fl_X = fl_Y \cdot \eta(*^v f)$$

* spoiler let $r: X \xrightarrow{\sim} {}^vX$

$$\kappa(e, X, r) := \text{tr}(fl_X \cdot \eta(r)) \in \Omega^2 \mathcal{C}$$

More flexible formula for fl.

Given $X \in \mathcal{C}$

✓ choose right and left duality data to define $\eta({}^v X)$ and $\eta(X)$

✓ choose right duality data for $\begin{matrix} {}^v X \boxtimes X \\ X \boxtimes {}^v X \end{matrix}$

✓ $\omega_L : ({}^v X \boxtimes X)^v \rightarrow X^v \boxtimes X$

$\omega_R : (X \boxtimes {}^v X)^v \rightarrow X \boxtimes X^v$

✓ mates for coev_{vX} , ev_{vX} , $\text{br}_{X, vX}$.

$$\begin{aligned} \eta({}^v X) &\doteq (\text{coev}_{vX})^v \circ (\text{br}_{X, vX})^v \circ (\text{ev}_{vX})^v \\ &\simeq (\text{coev}_{vX})^v \circ \underbrace{\omega_L^{-1} \circ \omega_L \circ \text{br}_{X, vX} \circ \omega_R^{-1} \circ \omega_R}_{\text{ev}_X \circ \text{br}_{X^v, X} \circ \text{coev}_X} \circ (\text{ev}_{vX})^v \\ &= \eta(X) \end{aligned}$$

S and T

$$\Omega S = \Omega T = \text{sVec}.$$

simple object (all invertible $\therefore \text{ev}_-$ and coev_- are isomorphisms)

✓ $\eta(I) \cong \text{id}_I \doteq I$

$$\left\{ \begin{array}{l} \eta(C) \cong e \\ \eta(M) \cong I \\ \eta(C \boxtimes M) \cong I \end{array} \right. \quad \left\{ \begin{array}{l} \text{br}_{C \boxtimes C} \cong e \boxtimes \text{id}_{C \boxtimes C} \\ \text{br}_{M \boxtimes M} \cong \text{id}_{M \boxtimes M} \end{array} \right.$$

$$\eta(e \boxtimes id_C) = \langle \underbrace{br_{id_C, e}}_1 \underbrace{br_{e, id_C}}_{-1} \rangle \underbrace{\eta(e)}_1 \underbrace{\eta(id_C)}_{-1} = -1 \cdot id_{\eta(C)}$$

$$\eta(e \boxtimes id_M) = \langle \underbrace{br_{id_M, e}}_{-1 \text{ (by defn. of } \eta M)} \underbrace{br_{e, id_M}}_{-1} \rangle \eta(e) \eta(id_M) = id_{\eta(M)}$$

$$\eta(e \boxtimes id_{C \boxtimes M}) = id_{\eta(C \boxtimes M)}$$

let ${}_I \mathcal{V}_C$ bimodule $\in \text{Hom}_{\Sigma \text{Vec}}(C, I)$

${}_C \mathcal{V}_I$ bimodule $\in \text{Hom}_{\Sigma \text{Vec}}(I, C)$

Fact ${}_I \mathcal{V}_C \circ {}_C \mathcal{V}_I \in \text{Hom}_{\Sigma \text{Vec}}(I, I) \cong \text{sVec}$
 $\parallel$
 C (as a super vector space)

$$\underbrace{id_I}_{\text{even}} \oplus \underbrace{e \boxtimes id_I}_{\text{odd}}$$

Consider $\eta({}_C \mathcal{V}_I \boxtimes id_M) : \eta(M) \xrightarrow[\textcircled{2}]{\textcircled{1}} \eta(C \boxtimes M) : \eta({}_I \mathcal{V}_C \boxtimes id_M)$

$$\textcircled{2} \circ \textcircled{1} = (id_I \oplus e) \boxtimes id_M$$

$$\eta(\textcircled{2}) \circ \eta(\textcircled{1}) = \eta(\textcircled{2} \circ \textcircled{1}) = \eta((id_I \oplus e) \boxtimes id_M) = 2 \cdot id_{\eta(M)} \quad \leftarrow \text{isomorphism } M \rightarrow M$$

similarly $\eta(\textcircled{1}) \circ \eta(\textcircled{2}) = \eta(\textcircled{1} \circ \textcircled{2})$ is an isomorphism

Therefore $\textcircled{1}$ and $\textcircled{2}$ are isomorphisms.
and $\eta(C \boxtimes M) \cong \eta(M) \cong I$.